Multi-timescale adaptive behavior in a dynamic patch foraging environment

Summary: Natural decisions unfold through continuous interaction between an organism and its environment. Foraging is an innate yet flexible behavior where animals repeatedly decide whether to exploit a depleting patch or leave to explore alternatives. We developed a head-fixed virtual reality patch foraging task that preserves the hierarchical and self-paced structure of natural behavior while allowing experimental control and access. Mice traversed a virtual linear track and made self-initiated choices to harvest probabilistic water rewards at odor-cued sites. Odors cued three patch types: high- and low-value patches whose reward probabilities depleted with each collected reward, and an unrewarded patch. Naive mice rapidly acquired the task, learning to use odor identity and reward outcomes to guide patch-leaving decisions in fewer than seven sessions on average. We used multi-timescale hidden markov models to identify hierarchically structured motifs in running velocity and respiration. Within a patch, run, approach, stop, and leave motifs occurred in sequence, and shifted as mice approached patch leaving, reflecting a latent decision variable. Individual mice had characteristic engagement state dynamics across sessions. Mice left both rewarded patch types at equivalent reward probabilities, consistent with the Marginal Value Theorem's (MVT, Charnov 1976) prediction to leave when local reward rate falls below the global average, but inconsistent with simpler heuristics such as reward counting or win-stay/lose-shift strategies. Independent manipulations of travel cost or global reward rate shifted patch-leaving decisions as predicted by MVT, demonstrating that mice use global reward rate to set local patch leaving thresholds. A Bayesian MVT (McNamara and Houston 1980) with two timescales of learning reproduced mouse behavior in steady-state and dynamic conditions. This task offers an avenue to investigate brain-wide circuit dynamics for adaptive behavior anchored in normative foraging theory.

Additional Detail:

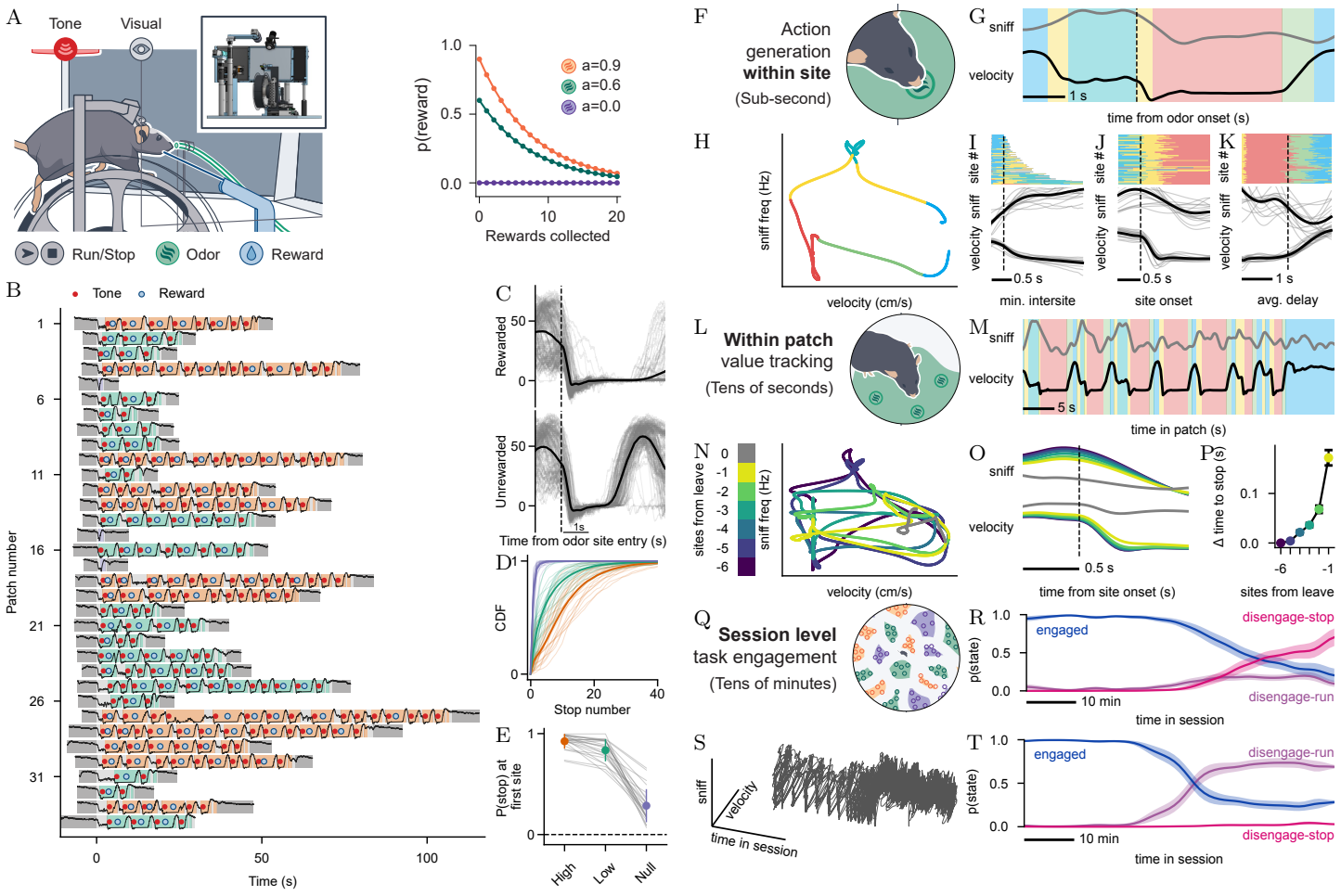


Figure 1: Multi-timescale structure of behavior in a virtual patch-foraging task. (A) Head-fixed mice run on a wheel through a virtual track with odor delivery. Between patches they run; at odor sites they choose to stop and harvest (a tone follows sustained stopping, then probabilistic reward) or run on. Patch types differ in initial reward probability, and depletes with each reward collected. (B) Example patches in session: velocity for every patch visit, with tones, rewards, and patch type (color). (C–E) Mice harvest rewards with stereotyped stop actions at odor site entries and remain stopped for longer in rewarded sites (C,E). Mice harvest more in high value patch types (D). (F,L,Q) Behavior is organized across nested timescales: sites (seconds), patches (tens of seconds to minutes), and sessions (tens of minutes). (G,H) Velocity and sniffing in anodor-site shaded by HMM states (G) and as trajectory in velocity–sniff frequency space (H). (I–K) HMM states (one row per site) and mean velocity aligned to approach state (I), odor site entry (J), and site leaving (K). (M,N) Velocity and sniffing in one patch. HMM states (M) and velocity–sniff trajectory colored by sites from leaving (N). (O,P) Action motifs shift as mice approach patch leaving: mice enter odor sites with decreased sniff frequency, increased velocity, (O), and slower reaction times (P; mean ± s.e.m., $n = 13$ mice). Colored by sites from leaving. (R,T) Session-averaged probability of three engagement states (engaged, disengage-run, disengage-stop; mean ± s.e.m. across sessions) for two mice, one disengaged by stopped (R, $n = 20$ sessions) the other disengaged by running (T, $n = 18$ sessions). (S) Example session as a trajectory in time–velocity–sniff space around patch entries.

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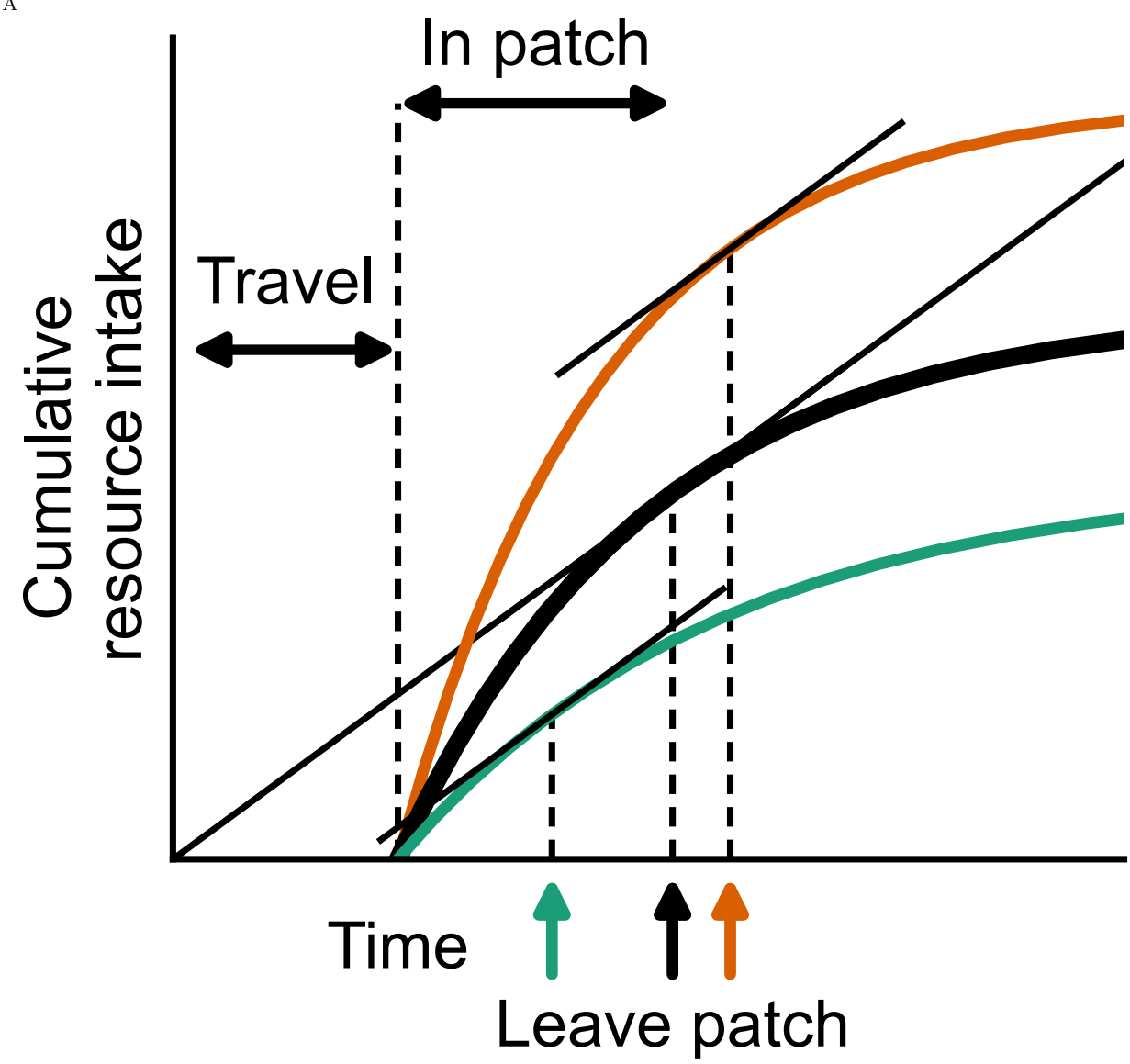


Figure 2: Patch-leaving behavior is consistent with a Bayesian formulation of the Marginal Value Theorem. (A) MVT predicts that an optimal forager leaves a patch when the instantaneous marginal reward rate falls below the global average reward rate, set by travel time between patches and within-patch reward statistics. (??) Reward probability at leaving is matched across patch types (ns), while rewards collected and number of reward omissions experienced differed significantly, ruling out simple heuristics such as reward counting or win-stay/loose switch. Lines connect individual animals ($N = 24$), Odor 1 (orange) and odor 2 (green) patch types. (??) Probability of leaving if the first odor site in a patch was unrewarded. Mice were more likely to leave the Low than the High patch after identical failure sequences, reflecting odor-cued prior expectations about patch quality. (??) Change in P(reward) at leaving relative to control as a function of inter-patch travel distance. Longer distances led mice to leave at lower reward probabilities, while shorter distances produced the opposite effect, consistent with MVT predictions. (??) Change in P(reward) at leaving relative to control as a function of inter-patch wheel friction. Increased friction raises the effective travel cost, leading mice to leave patches at lower reward probabilities. (??,??) Reward probability depletion curves for the transition to a higher (??) and lower (??) global reward rate. In (??), odor 2 reward statistics were increased to match those of odor 1; in (??), odor 1 reward statistics were decreased to match those of odor 2. Solid lines show the new reward statistics; faded lines show the original conditions. (??) Change in P(reward) at leaving relative to control under increased global reward rate conditions in mice. Both patch types shifted to higher reward probabilities, meaning mice left patches earlier. To achieve this, rewards collected shifted in opposite directions across patch types. This demonstrates cross-patch generalization consistent with global reward rate integration. (??) Opposite to (??); under decreased global reward rate, both patch types shifted to lower reward probabilities, with rewards collected shifting in opposite directions across patch types. (??) Bayesian MVT model schematic. The agent maintains a separate posterior over reward probability q for each odor-cued patch type, updated after each binary outcome via Bayes' rule with a forgetting parameter $\delta < 1$ that discounts accumulated prior beliefs at each update step. At each odor site, a sample $\tilde{q}$ is drawn from the current posterior; the agent leaves if the implied local reward rate falls below a running estimate of the global reward rate $\bar{R}$. (??–??) Bayesian MVT model simulations reproducing the mouse results in (??–??): reward probability at leaving and rewards collected (??), probability of leaving after early consecutive omissions (??), change in P(reward) at leaving with inter-patch distance (??), and inter-patch friction (??). (??,??) Model simulation depletion curves for the transition to a higher (??) and lower (??) global reward rate, corresponding to (??) and (??). (??,??) Same as (??) and (??) for Bayesian MVT model simulations, reproducing the cross-patch generalization observed in mice. Significance (paired t-tests across subjects): ns, not significant; $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.